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Calculation of Spin Correlations in Two-Dimensional Ising Systems from One-Dimensional Kinetic Models

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Kinetic spin models of the type first introduced by Glauber are considered with the most general choice of transition rates. It is shown that their time evolution operator can be related to the transfer matrices of certain two-dimensional Ising lattices. This allows an exact calculation of spin correlation functions at certain temperatures. Specifically we show that for the triangular antiferromagnet this special temperature corresponds to the disorder point. For the Hamiltonian version of the ANNNI model the corresponding result restricts in an important way the possible structure of the phase diagram.

I. Introduction

It is a well-known fact that all thermodynamic quantities of two-dimensional Ising lattices with short-range interactions can be derived from a one-dimensional operator, the transfer matrix V. This operator can always be expressed in terms of Pauli matrices, but its structure is normally complicated, involving products of exponentials which do not commute. A simplification occurs in two cases. Either one writes the operator V in the form $\exp(-\mathscr{H})$ with a new operator $\mathscr{H}$ (Hamiltonian limit) or one finds an operator $\hat{\mathscr{H}}$ which commutes with V and so has the same eigenfunctions. The problem the reduces to the study of $\mathscr{H}$ or $\hat{\mathscr{H}}$, which often are hermitian and therefore describe certain one-dimensional spin one-half problems. For example, for the Baxter model [1] $\hat{\mathscr{H}}$ is the XYZ Heisenberg Hamiltonian [2] and this correspondence was used to determine the asymptotic behaviour of correlation functions [3].

There is another type of problem which is conceptually closely related, the kinetic Ising model first introduced by Glauber [4]. It was constructed to study the approach of the spin systems towards the thermodynamic equilibrium state. Specifically, one writes down a master equation which governs the time evolution of the probability $p(\sigma, t)$ of finding a certain spin configuration σ at time t. This master equation can again be written in such a form that the time evolution operator T is expressed in terms of Pauli matrices, is hermitian and thus describes, for a linear chain, a one-dimensional quantum mechanical spin one-half problem.

In the present paper we shall explore the connections between these two types of problem and show what can be learned in that way. We shall mainly use a specific kinetic model where only flips of one spin at a time are allowed. The flipping rates will depend on the configurations of the neighbouring spins since we assume a nearest neighbour interaction along the chain. We shall consider the most general choice of the elementary rates. Our kinetic model then contains two free rate parameters. By choosing them in a proper way we then make contact with two different two-dimensional Ising problems, the triangular antiferromagnet and the Ising

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model with near neighbour and, in one direction, competing next nearest neighbour interactions (ANNNI model). To be precise, we show that $T=\mathscr{H}$ or $T=\hat{\mathscr{H}}$ along a certain line in the phase diagram of the two-dimensional model.

A consequence of such a relationship is that one can use the kinetic model to calculate e.g. spin correlation functions in a certain direction of the two-dimensional lattice. This calculation is then trivial, because the kinetic model is constructed in such a way that the (one-dimensional) Boltzmann distribution is the "lowest", i.e. stationary state. Therefore the correlation functions, calculated with this distribution, have strictly one-dimensional character. We shall call the line in the phase diagram of the $2D$ model where this happens, the "one-dimensional line (ODL)".

For the triangular antiferromagnet this effect was found before by Stephenson [5, 6] and studied in considerable detail. For the ANNNI model (in the Hamiltonian limit) the result is new and leads to an important restriction on the possible phase structure of this model, which so far is still being investigated.

The layout of the paper is as follows. The kinetic model with the Pauli operator representation of T is introduced in Sect. II and related to the antiferromagnet and to the ANNNI model in Sects. III and IV, respectively. Section V shows how the results for the latter case can also be derived directly without reference to the kinetic model. Section VI indicates some further correspondences which can be made between the ANNNI model and kinetic models. In Sect. VII we draw the conclusions.

II. Kinetic Spin Chain

Consider a linear chain of Ising spins $\sigma_n=\pm 1$, $n=1,2\ldots,N$ which interact via a nearest neighbour exchange coupling J. Let $\boldsymbol{\sigma}=\{\sigma_1,\sigma_2\ldots\sigma_N\}$ denote a spin configuration of the chain. Its energy is $H(\boldsymbol{\sigma})=-J\sum_n \sigma_n\sigma_{n+1}$. We also assume cyclic boundary conditions, $\sigma_{N+1}=\sigma_1$. If $p(\boldsymbol{\sigma},t)$ denotes the probability of finding the configuration $\boldsymbol{\sigma}$ at time t, the master equation is

$$\frac{\partial}{\partial t}p(\boldsymbol{\sigma},t)=-\sum_{\boldsymbol{\sigma}'} w(\boldsymbol{\sigma}\to\boldsymbol{\sigma}')\,p(\boldsymbol{\sigma},t)+\sum_{\boldsymbol{\sigma}'} w(\boldsymbol{\sigma}'\to\boldsymbol{\sigma})\,p(\boldsymbol{\sigma}',t). \tag{2.1}$$

The transition probabilities $w(\boldsymbol{\sigma}\to\boldsymbol{\sigma}')$, have now to be specified. We assume that w is a superposition of

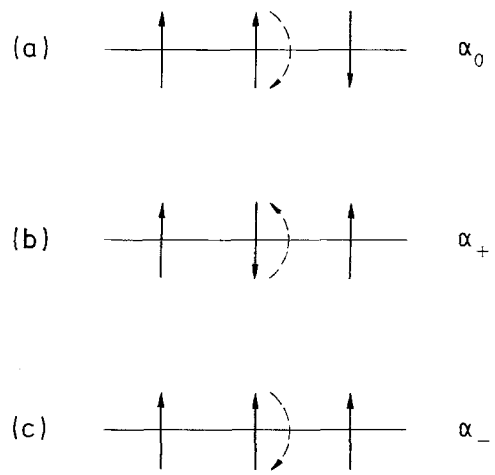

Fig. 1. The elementary spin-flip processes in the one-dimensional kinetic model and their rates. Processes **b** and **c** are connected by detailed balance

one-spin-flip terms. Assuming up-down and left-right symmetry there are only three different flip processes to consider, which are shown in Fig. 1. If we assume detailed balance, the rates of process (*b*) and (*c*) are connected by $\alpha_-=e^{-4K}\alpha_+$ where $K=\beta J$ and $\beta=(k_B\mathrm{T})^{-1}$ is related to the temperature.

We are left, therefore, with two independent rate constants α_0 and α_+, say. Such a model is more general than Glauber's original one and has been studied previously [7, 8, 9, 10].

The master equation can be written in the form

$$\frac{\partial}{\partial t}\tilde{p}(\boldsymbol{\sigma},t)=-\sum_{\boldsymbol{\sigma}'} T(\boldsymbol{\sigma},\boldsymbol{\sigma}')\,\tilde{p}(\boldsymbol{\sigma}',t) \tag{2.2}$$

where the reduced probability $\tilde{p}$ is defined as

$$\tilde{p}(\boldsymbol{\sigma},t)=p_0(\boldsymbol{\sigma})^{-1/2}\cdot p(\boldsymbol{\sigma},t) \tag{2.3}$$

and

$$p_0(\boldsymbol{\sigma})=\mathrm{const}\cdot\exp\{-\beta H(\boldsymbol{\sigma})\} \tag{2.4}$$

is the equilibrium distribution function. The form (2.2) has the advantage that $T(\boldsymbol{\sigma},\boldsymbol{\sigma}')$ is explicitely symmetric.

The operator $T(\boldsymbol{\sigma},\boldsymbol{\sigma}')$ can now be expressed in terms of Pauli matrices $\sigma_n^x,\sigma_n^y,\sigma_n^z$, $n=1,2\ldots N$. This calculation was already performed in [9] so that we just quote the final result:

$$T=-\sum_{n=1}^{N}(A\sigma_n^x+B\sigma_{n-1}^z\sigma_n^x\sigma_{n+1}^z+C\sigma_n^z\sigma_{n+1}^z-D\sigma_n^z\sigma_{n+2}^z-E) \tag{2.5}$$

where with $\bar{\alpha}=\alpha_+\,e^{-2K}$

$$A = \tfrac{1}{2}(\bar{\alpha} + \alpha_0)$$
$$B = \tfrac{1}{2}(\bar{\alpha} - \alpha_0)$$
$$C = \bar{\alpha} \cdot sh\, 2K$$
$$D = \tfrac{1}{2}(\bar{\alpha}\, ch\, 2K - \alpha_0)$$
$$E = \tfrac{1}{2}(\bar{\alpha}\, ch\, 2K + \alpha_0). \tag{2.6}$$

Equation (2.5) gives a very transparent description of the problem in question. For example, all operators except the term $\sigma_n^z \sigma_{n+2}^z$ are expressible as bilinear forms of fermion operators. To see this one interchanges σ^x and σ^z and then uses the Jordan-Wigner transformation. Now Glauber's original choice was [4]

$$\alpha_0 = ch\, 2K \cdot \bar{\alpha} \tag{2.7}$$

which implies $D=0$, hence the exceptional term was absent. This enabled Felderhof [11] to carry out the calculation of time-dependent correlation functions purely in terms of fermions, although the algebra was rather involved.

On the other hand, two recent papers [9, 10] showed that for the choice

$$\alpha_0 = e^{-2K} \cdot \bar{\alpha} \tag{2.8}$$

which leaves all coefficients different from zero, but gives e.g. $C=D$, the spin susceptibility $\chi(q=0,\omega)$ still can be calculated exactly and is the sum of two Lorentzians. This is an interesting example of an interacting Fermi system which admits a partial solution. In the following we shall consider two other choices of the coefficients appearing in (2.5).

Finally, let us mention that the function $p_0(\sigma)^{1/2}$ is an eigenstate of T with eigenvalue zero since we used detailed balance. It also can be shown [12] that with detailed balance and positive rates all other eigenvalues of T are larger than zero so that $p_0^{1/2}$ is in fact the lowest state.

III. The Triangular Ising Model

Let us consider a triangular Ising lattice as shown in Fig. 2 where the bonds in all three directions are different. The transfer matrix between rows n and $(n+1)$ has the same structure as for a quadratic lattice with diagonal interaction J_2. This can be seen by distorting the lattice and is shown in Fig. 2.

Call the symmetric transfer matrix corresponding to Fig. 2 $V(K_1, K_2, K_3)$ where $K_i = \beta J_i$ and β is again the inverse temperature (of the two-dimensional problem now.) While V itself is somewhat complicated, it was shown by Stephen and Mittag [13]

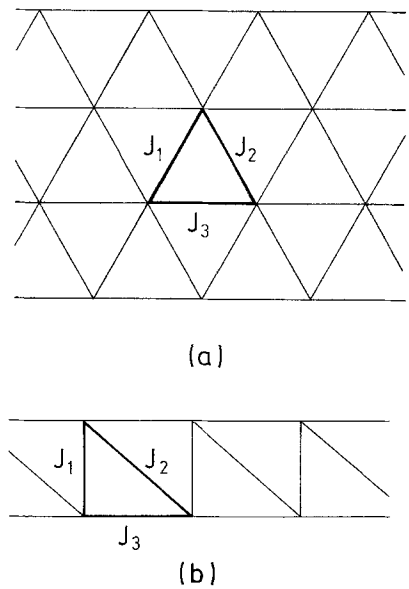

Fig. 2. a The triangular Ising lattice and its coupling constants. **b** Two consecutive rows after a distortion which shows the relation to a quadratic lattice

that it commutes with the operator

$$\hat{\mathscr{H}}_\Delta = -\sum_{n=1}^{N} (\sigma_n^x + k_2\, \sigma_{n-1}^z\, \sigma_n^x\, \sigma_{n+1}^z + k_1\, \sigma_n^z\, \sigma_{n+1}^z) \tag{3.1}$$

where, with $S_i = sh\, 2K_i$, $C_i = ch\, 2K_i$

$$k_1 = 2(S_1\, S_2\, C_3 + C_1\, C_2\, S_3)/(1 + C_3)$$
$$k_2 = -[S_3/(1 + C_3)]^2. \tag{3.2}$$

Comparing with (2.5) we see that $\hat{\mathscr{H}}_\Delta$ has the same form as T provided $D=0$ and the coefficients k_2 and k_1 are equated to B/A and C/A. $D=0$ means that V is related to the original Glauber model, i.e. to a free fermion problem, as it should be. Putting $D=0$ we obtain

$$T = -\bar{\alpha}\, ch^2 K \cdot \sum_n (\sigma_n^x - th^2 K\, \sigma_{n-1}^z\, \sigma_n^x\, \sigma_{n+1}^z$$
$$+ 2th\, K\, \sigma_n^z\, \sigma_{n+1}^z - 1 - th^2 K). \tag{3.3}$$

This coincides - up to a multiplicative and an additive constant, which do not affect the commutation relation - with $\hat{\mathscr{H}}_\Delta$ of (3.1) if

$$k_1 = 2th\, K$$
$$k_2 = -th^2 K. \tag{3.4}$$

This means that the two quantities k_1, k_2 are related by

$$k_1^2 = -4k_2. \tag{3.5}$$

(3.5) is a constraint on the quantities K_i or, for fixed coupling constants, on the temperature of the triangular lattice. Using (3.2) it may be rewritten as

$$S_1 S_2 C_3 + C_1 C_2 S_3 = \pm S_3. \tag{3.6}$$

For a ferromagnet with all $K_i > 0$ this can never be fulfilled since the left hand side of (3.6) is always larger than the right hand side. For an antiferromagnet with $K_i < 0$, it can be fulfilled if one chooses the minus sign.

We therefore have the following result: There exists a certain temperature, determined by (3.6), such that V commutes with the operator T of Glauber's model. The coupling constant K of the kinetic model is then given by (3.4).

If we now make the additional assumption that the state corresponding to the largest eigenvalue of V is the ground state $p_0(\sigma)^{1/2}$ of T, we are able to calculate any spin correlation function in the triangular lattice along the direction 3.

$$\begin{aligned}\langle \sigma_l \sigma_m \sigma_n \ldots \rangle_\Delta &= \langle 0 | \sigma_l \sigma_m \sigma_n \ldots | 0 \rangle \\ &= \sum_\sigma \sigma_l \sigma_m \sigma_n \ldots p_0(\sigma) \\ &= \frac{1}{Z} \sum_\sigma \sigma_l \sigma_m \sigma_n \ldots \exp(K \sum_n \sigma_n \sigma_{n+1}).\end{aligned} \tag{3.7}$$

Specifically, the spin-spin correlation function is

$$\langle \sigma_l \sigma_{l+n} \rangle = |th\, K_3|^n, \qquad n \geqq 0. \tag{3.8}$$

The result (3.8) has been obtained before by Stephenson [5] who discussed the spin correlation functions of the triangular lattice for *all* temperatures. He noticed that at a certain temperature, which he called the disorder temperature T_D, the correlations are purely one-dimensional in character. His defining equation for the disorder temperature is precisely (3.6).

In a phase diagram (temperature vs. the ratio of coupling constants J_3/J_2) the disorder temperatures lie on a line which runs through the region without long-range order. It does not separate different phases of the system but the character of the correlation functions changes at the disorder point. For $T < T_D$ the correlation function decays monotonically to zero (for $J_3 < J_2 < J_1 < 0$ which we assumed tacitly) while for $T > T_D$ the decay is oscillatory with a wave vector which depends on the temperature. In addition, the correlation length $\xi(T)$ has a cusp a $T = T_D$ [6].

Disorder points have been found in various other systems such as the Union-Jack lattice [14, 15], the Kagomé lattice [16], a one-dimensional model of classical spins [17], classical hard core fluids [18] or the one-dimensional ANNNI model [19, 20]. The nature of the correlation functions at the disorder temperature can, however, vary from system to system. In this respect the Kagomé lattice is an extreme case since there at T_D the correlation between spins in a certain direction is identically zero. An elegant physical interpretation as a decoration effect was given by Fradkin and Eggarter [21]: Direct bonds and decorated bonds together determine the effective coupling which is temperature dependent and can vanish in certain directions at certain temperatures. In this way a two-dimensional system can behave like a one-dimensional or even a zero-dimensional one. Our approach represents the mathematical formulation of this idea. By showing that T and V commute we prove directly (under the forementioned assumption on the eigenfunctions) that the two-dimensional system behaves in a one-dimensional way.

IV. The ANNNI Model

This is the simplest model for studying the effect of competing interactions. The second nearest neighbour interaction J_2 operates only in one spatial direction. The model was used to discuss the magnetism of CeSb and studied extensively by Bak and von Boehm [22] and Selke and Fisher [23, 24]. The mean field theory [22] as well as a low temperature expansion [24] show a very rich phase diagram with a sequence of periodic structures. A first approximate treatment in two dimensions is due to Villain and Bak [25]. Ruján [26] showed e.g. how to relate it to a dimerized eight vertex model with fields. Numerical studies include high and low temperature expansions [27] and calculations on systems of finite size [27–29, 35].

The row-to-row transfer matrix of the model is easily written down and is given by

$$V = \exp(K_0^* \sum_n \sigma_n^x) \cdot \exp(\sum_n [K_1 \sigma_n^z \sigma_{n+1}^z + K_2 \sigma_n^z \sigma_{n+2}^z]) \tag{4.1}$$

where $th\, K_0^* = e^{-2K_0}$, $K_0 = \beta J_0$ and J_0 is the coupling in the vertical direction. Since V is difficult to handle in this form, let us consider the Hamiltonian limit which consists of pulling the exponentials together. Formally this is correct, if $K_0^*, K_1 K_2 \ll 1$ which corresponds to strong coupling in the vertical and weak coupling in the horizontal direction. Nevertheless this Hamiltonian version of the ANNNI model is interesting in its own right for all values of the parameters although it should be recognized that some features of its phase diagram may be different from

the original model. Writing $V=\exp(-K_1\mathscr{H})$ we have

$$\mathscr{H}=-\sum_{n=1}^{N}(y\sigma_n^x+\sigma_n^z\sigma_{n+1}^z+\kappa\sigma_n^z\sigma_{n+2}^z) \tag{4.2}$$

with $y=K_0^*/K_1$, $\kappa=K_2/K_1$. To be specific we will choose $K_1>0$, $K_2<0$, i.a. antiferromagnetic coupling to the second neighbours.
Comparing $\mathscr{H}$ with T we see that both can agree this time if $B=0$, i.e.

$$\alpha_0=\bar{\alpha}. \tag{4.3}$$

This gives

$$T=-\sum_n(\bar{\alpha}\sigma_n^x+\bar{\alpha}\,sh\,2K\cdot\sigma_n^z\sigma_{n+1}^z$$
$$-\bar{\alpha}\,sh^2K\,\sigma_n^z\sigma_{n+2}^z-\bar{\alpha}\,ch^2K). \tag{4.4}$$

Equations (4.2) and (4.4) are compatible if $y=\bar{\alpha}$, $1=\bar{\alpha}\,sh\,2K$, $\kappa=-\bar{\alpha}\,sh^2K$ or, eliminating $\bar{\alpha}$, if

$$y=\kappa-\frac{1}{4\kappa}. \tag{4.5}$$

If this relation is fulfilled, we have exactly

$$\mathscr{H}=T+N\cdot\frac{1}{4\kappa} \tag{4.6}$$

and we also know the ground state of both T and $\mathscr{H}$. Note that the two models can be connected *only* if $\kappa<0$, i.e. if the interactions J_1 and J_2 are of different sign. This fact has also been noticed by Siggia [8]. Along the line given by (4.5) the spin correlation functions in the horizontal direction again have one-dimensional character:

$$\langle\sigma_l\sigma_{l+n}\rangle=(-2\kappa)^n,\qquad n\geqq 0. \tag{4.7}$$

Since $\kappa<0$ this is a purely exponential monotonous decay with correlation length $\xi^{-1}=-\ln(-2\kappa)$.
In Fig. 3 we have plotted the one-dimensional line together with the approximate phase boundary of the ferromagnetic state

$$1+2\kappa=y+\frac{\kappa}{2(1+\kappa)}\cdot y^2 \tag{4.8}$$

which follows from the vanishing of the mass gap obtained to second order in y. As one approaches the point $\kappa=-0.5$, $y=0$ both lines merge. They have the same slope and agree to order y^2. By the way, if one deletes the y^2-term in (4.8) one obtains a slightly different phase boundary which was found previously with a method due to Müller-Hartmann and Zittartz [20].

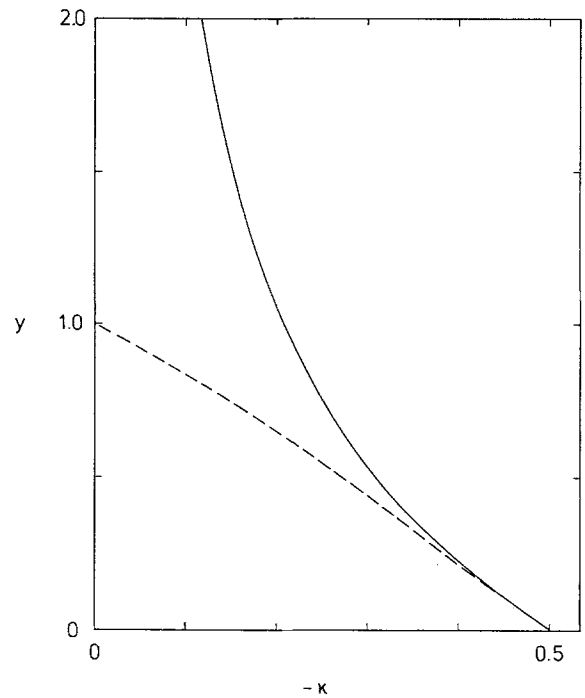

Fig. 3. The one-dimensional line, (4.5), (full) and the ferromagnetic phase boundary, (4.8), (dashed) for the ANNNI model

The significance of the ODL is that it excludes – where it runs – any ordered phase or even a "floating phase" with algebraic decay of the correlation functions. Earlier phase diagrams [20] gave such phases immediately adjacent to the ferromagnetic phase. Insofar as the Hamiltonian limit is concerned they are therefore incompatible with our exact result.
In contrast to the triangular Ising model the precise relationship between the ODL and the disorder point is not known at present. The ANNNI system must have a disorder line since the correlation functions have ferromagnetic character for $\kappa=0$ (or for $\kappa>-0.5$ if $y=0$) and antiferromagnetic character for $|\kappa|$ and y large (or for $\kappa<-0.5$ if $y=0$). It is clear that the disorder line cannot be below the ODL in the $\kappa-y$ plane since the correlations have ferromagnetic character on the ODL. Both lines pass through the point $\kappa=-0.5$, $y=0$. Furthermore the Lifschitz point [20] at which the disorder line meets the critical line must be at $\kappa=-0.5$, $y=0$, too.
From (4.6) one can also deduce the free energy along the ODL

$$\frac{F}{N\cdot J_1}=\frac{1}{4\kappa}. \tag{4.9}$$

This is the same formula for all sizes of the system and has been checked by finite size calculations [28].

V. Alternative Treatment of the ANNNI Problem

The results of the preceding section show that on the ODL the operator $\mathscr{H}$ has a very simple ground state wave function. Once this has been established, it can be verified in a more direct way. For that purpose it is convenient to work with dual variables. It will turn out that in this representation the ground state wave function has the form of a simple product.

In the dual transformation one introduces new variables τ_n via $\tau_n = \sigma_n \sigma_{n+1}$. The Pauli operators transform according to

$$\begin{aligned} \sigma_n^z \sigma_{n+1}^z &= \tau_n^z \\ \sigma_n^x &= \tau_{n-1}^x \tau_n^x \end{aligned} \tag{5.1}$$

which leads to the dual Hamiltonian

$$\mathscr{H}_D = -\sum_n (y \tau_n^x \tau_{n+1}^x + \tau_n^z + \kappa \tau_n^z \tau_{n+1}^z). \tag{5.2}$$

The advantage of $\mathscr{H}_D$ over $\mathscr{H}$ is that it does not contain next nearest neighbour interactions any more.

Next, we rotate all spins by an angle θ around the y axis in spin space. This transformation can be generated by the unitary operator

$$U = \prod_n \exp\left(-i \frac{\theta}{2} \tau_n^y\right). \tag{5.3}$$

The transformed Hamiltonian is given by

$$U \mathscr{H}_D U^{-1} = \mathscr{H}_0 + \mathscr{H}_1 + \mathscr{H}_2 \tag{5.4}$$

with

$$\begin{aligned} \mathscr{H}_0 &= -\sum_n \{c \tau_n^z + (y s^2 + \kappa c^2) \tau_n^z \tau_{n+1}^z\} \\ \mathscr{H}_1 &= -\sum_n s \{\tau_n^x [\tfrac{1}{2} + c(\kappa - y) \tau_{n+1}^z] \\ &\quad + \tau_{n+1}^x [\tfrac{1}{2} + c(\kappa - y) \tau_n^z]\} \\ \mathscr{H}_2 &= -\sum_n \{\kappa s^2 + y c^2\} \tau_n^x \tau_{n+1}^x \end{aligned} \tag{5.5}$$

where $c = \cos\theta$, $s = \sin\theta$. The ground state of $\mathscr{H}_0$ is the ferromagnetic state with all spins up (for $\kappa > -0.5$) and its energy can be minimized by choosing $c = [2(y-\kappa)]^{-1}$. With this choice $\mathscr{H}_1$ becomes

$$\mathscr{H}_1 = -\frac{S}{2} \sum_n \{\tau_n^x (1 - \tau_{n+1}^z) + \tau_{n+1}^x (1 - \tau_n^z)\} \tag{5.6}$$

and the ferromagnetic state is also an eigenstate of $\mathscr{H}_1$ with eigenvalue zero. Finally

$$\mathscr{H}_2 = -\sum_n \left\{\kappa + \frac{1}{4} \frac{1}{y - \kappa}\right\} \tau_n^x \tau_{n+1}^x \tag{5.7}$$

can be made to vanish identically by choosing y on the ODL. We therefore have shown that the state with all τ-spins up is an eigenstate of the rotated dual Hamiltonian. For its energy we find

$$\frac{E_0}{N} = \frac{1}{4\kappa} \tag{5.8}$$

which is the same result as in (4.9). Our state therefore has to be related to the distribution function $p_0(\boldsymbol{\sigma})^{1/2}$ which is the lowest state of the original model. To find the precise relation let us consider a symmetry property of $\mathscr{H}_D$ for a moment. It is easy to see that $\mathscr{H}_D$ commutes with the parity operator

$$P_D = \prod_n \tau_n^z \tag{5.9}$$

so that both have common eigenfunctions. Now, rotating the function we found back to the initial dual representation, it becomes

$$|\psi(\theta)\rangle = \prod_n \left[\cos\frac{\theta}{2} \cdot |+, n\rangle - \sin\frac{\theta}{2} \cdot |-, n\rangle\right] \tag{5.10}$$

where $|\pm, n\rangle$ are the eigenstates of the spin operator τ_n^z at site n. Operating with P_D on $|\psi(\theta)\rangle$ one finds that $|\psi(\theta)\rangle$ is *not* an eigenfunction. Therefore there has to be another function which is degenerate with $|\psi(\theta)\rangle$. It is not difficult to see that this function is $|\psi(-\theta)\rangle$. Then the linear combinations

$$\begin{aligned} |\psi_\pm\rangle &= A_\pm \cdot [|\psi(\theta)\rangle \pm |\psi(-\theta)\rangle] \\ A_\pm &= [2(1 \pm c^N)]^{-1/2} \end{aligned} \tag{5.11}$$

are normalized eigenfunctions of $\mathscr{H}_D$ and P_D with parity ± 1.

This result seems to be strange. The ground state of $\mathscr{H}$ is non-degenerate on the ODL while the ground state of $\mathscr{H}_D$ was found to be doubly degenerate. Looking carefully at the dual transformation one finds, however, that only the even parity states of $\mathscr{H}_D$ have relevance for the original problem if the σ_n (as we assumed) obey periodic boundary conditions. Therefore it is the state $|\psi_+\rangle$ which corresponds exactly to the state $p_0(\boldsymbol{\sigma})^{1/2}$. This can be checked, for example, by calculating the expection value

$$\langle \psi_+ | \tau_n^z | \psi_+ \rangle = \frac{c + c^N}{1 + c^N}. \tag{5.12}$$

This has to be the same as the nearest neighbour correlation function

$$\langle 0 | \sigma_n^z \sigma_{n+1}^z | 0 \rangle = \sum_{\boldsymbol{\sigma}} \sigma_n^z \sigma_{n+1}^z \, p_0(\boldsymbol{\sigma}) \tag{5.13}$$

in the original variables. Inserting $p_0(\sigma)$, (2.4), it is easy to calculate this and with the connections $c = th\,K = -2\kappa$ on the ODL one finds indeed the same result, (5.12). The simple form of the function $|\psi_+\rangle$ allows, of course, the calculation of arbitrary correlation functions on the ODL.

If we consider the dual model as a problem in itself there is, of course, no reason to leave out the odd parity states. With our wave functions we then are able to calculate e.g. the "dual order parameter"

$$\langle\psi_-|\tau_n^x|\psi_+\rangle = -\frac{s}{(1-c^{2N})^{1/2}} \tag{5.14}$$

along the ODL. Furthermore, it is possible to see how the degenerate ground state splits if one goes *off* the ODL. To first order in the deviation $\delta y = y - y_{\mathrm{ODL}}$ one obtains

$$\frac{1}{N}E_0^{\pm} = \frac{1}{4\kappa} - \delta y \cdot \frac{1-4\kappa^2}{1\pm(-2\kappa)^N}. \tag{5.15}$$

This result shows that off the ODL the two energies are, for large N, exponentially close to each other. This is the usual behavior for a system with long range order (note that the dual model is ordered for large values of y). By contrast, the ground state is degenerate *on* the ODL for *all* values of N. This behaviour has also been seen in numerical calculations on small systems [29, 28]. Finally, note that the phase boundary and the ODL come very close to each other as κ approaches the value -0.5 (Fig. 3). The exact degeneracy at the ODL therefore might cause problems when one derives critical properties of $\mathscr{H}_D$ from series expansions. Such difficulties have been noted by Barber and Duxbury [27].

VI. Alternative Connections

The dual transformation of (5.1) can be performed in a slightly different way. One first doubles the Ising lattice in the horizontal direction by fitting in another lattice in an interlacing fashion. A following dual transformation gives

$$\mathscr{H}_{DD} = -\sum_n (y\sigma_n^x\sigma_{n+1}^x + \sigma_n^z\sigma_{n+1}^z + \kappa\sigma_n^z\sigma_{n+1}^z\sigma_{n+2}^z\sigma_{n+3}^z). \tag{6.1}$$

This operator contains now a four-spin coupling term.

The question arises whether $\mathscr{H}_{DD}$ can also be related to a kinetic spin model. This is in fact the case, but the kinetic model needed has to have *two* spin-flips at a time at neighbouring sites. Models of this type

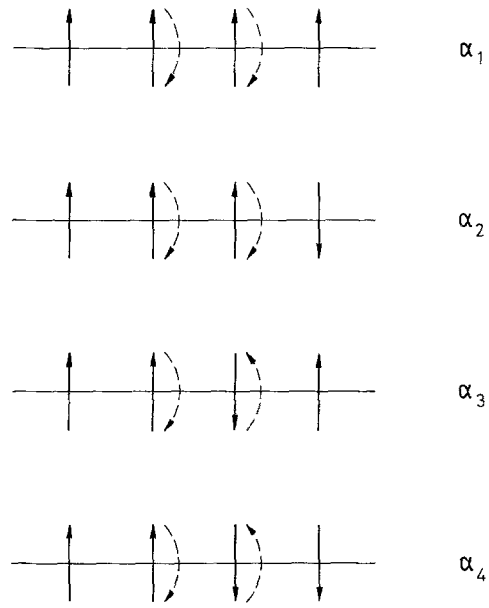

Fig. 4. The four independent spin-flip processes in the two-flip model and their rates

have been discussed by Hilhorst et al. [30] and Haake and Thol [31]. The most general model contains four independent rate parameters and the corresponding processes are shown in Fig. 4. In the notation of [31] one has

$$\begin{aligned}
\alpha_1 &= \Gamma(1+\alpha+\beta+\mu)(1-th\,2K)\\
\alpha_2 &= \Gamma(1+\alpha-\beta-\mu)\\
\alpha_3 &= \Gamma(1-\alpha+\beta-\mu)\\
\alpha_4 &= \Gamma(1-\alpha-\beta+\mu)(1-th\,2K).
\end{aligned} \tag{6.2}$$

A tedious but straightforward calculation gives for T:

$$\begin{aligned}
T = -\sum_n \{&J_x\sigma_n^x\sigma_{n+1}^x + J_y\sigma_n^y\sigma_{n+1}^y + J_z\sigma_n^z\sigma_{n+1}^z\\
&+\sigma_{n-1}^z[J_{xz}\sigma_n^x\sigma_{n+1}^x + J_{yz}\sigma_n^y\sigma_{n+1}^y + J_{zz}\sigma_n^z\sigma_{n+1}^z]\,\sigma_{n+2}^z\\
&+J_2\sigma_n^z\sigma_{n+2}^z + J_4\sigma_{n-1}^z\sigma_{n+2}^z + C\}
\end{aligned} \tag{6.3}$$

with the parameters

$$\begin{aligned}
J_x &= \Gamma\cdot[(1+\mu)+ch\,2K\cdot(1-\mu)]/2ch\,2K\\
J_y &= \Gamma[(\alpha+\beta)+ch\,2K\cdot(\alpha-\beta)]/2ch\,2K\\
J_z &= -\Gamma[\alpha\,ch\,2K - sh\,2K\cdot(1+\mu)]/ch\,2K\\
J_{xz} &= \Gamma[(\alpha+\beta)-ch\,2K\cdot(\alpha-\beta)]/2ch\,2K\\
J_{yz} &= -\Gamma[(1+\mu)-ch\,2K\cdot(1-\mu)]/2ch\,2K\\
J_{zz} &= -\Gamma\cdot\mu\\
J_2 &= \Gamma(\alpha+\beta)\cdot th\,2K\\
J_4 &= -\Gamma\cdot\beta\\
C &= -\Gamma.
\end{aligned} \tag{6.4}$$

Let us remark first that T contains processes which conserve the total spin (rates α_3 and α_4) and processes which do not, i.e. where the two spins are both flipped up or down at the same time. If one suppresses these latter processes by setting $\alpha_1=\alpha_2=0$ or $\alpha=-1$, $\beta=-\mu$ one obtains a Kawasaki model [32] where the total magnetization is a constant of the motion. In this case T simplifies considerably. In particular for $K=0$ one obtains

$$T=-\Gamma\sum_n\{(\vec{\sigma}_n\cdot\vec{\sigma}_{n+1}-1)+\beta\sigma^z_{n-1}(\vec{\sigma}_n\cdot\vec{\sigma}_{n+1}-1)\,\sigma^z_{n+2}\}. \tag{6.5}$$

For $\beta=0$ this is the isotropic Heisenberg model, which therefore can be used to study conduction processes if one interprets $n_l=(1+\sigma^z_l)/2$ as a particle number [33, 34].

A connection to the ANNNI model can be made if one chooses

$$J_y=J_{xz}=J_{yz}=J_2=J_4=0.$$

These five relations can indeed be satisfied by setting $\alpha=\beta=0$, $\mu=th^2K$. The resulting T is

$$T=-\Gamma\cdot sech^2K\cdot\sum_n\{\sigma^x_n\sigma^x_{n+1}+sh^2K\,\sigma^z_n\sigma^z_{n+1}-sh^2K\cdot\sigma^z_n\sigma^z_{n+1}\sigma^z_{n+2}\sigma^z_{n+3}\}. \tag{6.6}$$

A comparison with (6.1) shows that this is $\mathscr{H}_{DD}$ if $y=(sh\,2K)^{-1}$, $\kappa=-(th\,K)/2$. This leads, therefore, to exactly the same relation (4.5) between y and κ as found before via the single-spin-flip model. While we have not found a new result, this shows another route to the one-dimensional line.

Finally, we want to show that even for the right hand side, $\kappa<-0.5$, of the phase diagram in Fig. 3 a connection of the ANNNI model to a spin model exists. To see this, first rotate the dual form $\mathscr{H}_D$ of (5.2) to obtain

$$\mathscr{H}'_D=-\sum_n(y\sigma^y_n\sigma^y_{n+1}+\sigma^x_n+\kappa\sigma^x_n\sigma^x_{n+1}). \tag{6.7}$$

Now perform a second dual transformation as in Equ. (5.1). The operator $\sigma^y_n\sigma^y_{n+1}$ then transforms into $-\sigma^x_{n-1}\sigma^z_n\sigma^x_{n+1}$ and, after another interchange of σ^x and $-\sigma^z$, one has

$$\mathscr{H}''_D=-\sum_n(y\sigma^z_{n-1}\sigma^x_n\sigma^z_{n+1}+\sigma^z_n\sigma^z_{n+1}+\kappa\sigma^z_n\sigma^z_{n+2}). \tag{6.8}$$

This can be compared with T, (2.5). In order that both operators be the same, one has to choose $A=0$ this time. This can, in fact, be achieved, but only with negative rates $\alpha_0=-\bar{\alpha}$. Accepting this for the moment, the two operators coincide if one has $\kappa<-0.5$ and

$$y=-\kappa+\frac{1}{4\kappa}. \tag{6.9}$$

This line starts from $y=0$ at $\kappa=-0.5$ and then rises in the right hand side of the phase diagram of Fig. 3. It is this region where antiferromagnetic ordering and possible floating phases are expected.

Although our correspondence is exact, the choice of negative elementary rates destroys one crucial property of the kinetic model: it can no longer be proved that the equilibrium distribution is the lowest state. In other words, the system might show exponentially growing solutions. A hint that the state $p_0^{1/2}$ is in fact no longer the ground state of $\mathscr{H}''_D$ comes from a calculation of the free energy. One obtains for $\kappa<-0.5$ the same formula (4.9) as before, but this result is incorrect in the soluble limit, $\kappa\to-\infty$.

VII. Conclusions

We have shown how the knowledge of the time evolution operator T for kinetic spin models can be used to deduce properties of two-dimensional Ising systems. This enabled us to draw conclusions on the triangular Ising lattice and on the ANNNI model. The considerations are exact up to one point. If we use the commutation property $[T, V]=0$, we have to make an assumption on the eigenfunctions of V. If we treat the ANNNI model, we have to go over to the Hamiltonian limit for V. However, we obtain as a result a very simple way to find special points or lines in the phase diagrams.

According to our method we could, of course, find only points where the two-dimensional system behaves in a very simple way: the correlation functions show one-dimensional, exponential decay. This is the consequence of using a kinetic model with nearest neighbour interactions. One could also start from a kinetic chain which has itself competing interactions. The correlation functions, while still one-dimensional in character, would then show more complicated, oscillatory behaviour. It could be that this behaviour again can be found at certain temperatures and in certain directions in two-dimensional systems.

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